CYS 331: WIRELESS SECURITY

Fully Elaborated Laboratory Manual

Unit 1: Wireless Technology

Lecturer: Dr Lungu Nelson

Mode: Self-guided student laboratory

Mark allocation: 3 marks

Use only personal devices, personal accounts, and personally controlled wireless environments

Course code	CYS 331
Course title	Wireless Security
Unit	Unit 1: Wireless Technology
Recommended duration	2 to 3 hours
Lab marks	3 marks

1. Scope and purpose

This lab is limited to the Unit 1 topics in the course outline: wireless technologies, wireless data networks, personal area networks, transmission media, WLAN standards, securing WLANs, countermeasures, and WEP. The student uses only a personal smartphone hotspot and a personal laptop to identify wireless settings, relate them to WLAN concepts, and improve the security of that personally controlled wireless environment.

This is a self-guided student laboratory. The student must use only personal devices, personal accounts, and a personally controlled wireless environment. No offensive activity, no unauthorised access, and no third-party data capture are permitted.

2. Tools and exact setup requirements

Tool / item	Exactly how to set it up	Why it is needed in this lab
Smartphone with hotspot capability	Use your own Android phone or iPhone. Android path is usually Settings > Network & Internet / Connections > Hotspot & tethering / Mobile Hotspot. iPhone path is Settings > Personal Hotspot. Before the lab starts, confirm that mobile data is on and that the phone battery is adequately charged.	The phone creates the authorised wireless network for the lab. No third-party network should be used.
Laptop with Wi-Fi	Use your own Windows laptop.	The laptop is used to view

	Confirm that Wi-Fi is turned on by selecting the network icon on the taskbar. If airplane mode is on, turn it off before you start.	available wireless networks, connect to the hotspot, and record evidence.
Screenshot method	On Windows 10/11 press Windows + Shift + S, then drag over the part of the screen you need. Open the captured image from the notification and save it into a folder named Unit1_Evidence. On Android use Power + Volume Down. On iPhone use Side Button + Volume Up.	Screenshots provide proof of what was actually observed and configured.
Command Prompt	On the Windows keyboard press Windows + R, type cmd, then press Enter. A black window titled Command Prompt will open. You will use the command netsh wlan show interfaces exactly as written.	This gives accurate values such as SSID, radio type, channel, authentication, and cipher.

3. Evidence the student must produce

- Labelled screenshots saved from the actual practical activity.
- A short table or matrix that records the required technical values.
- A concise explanation of what each important setting or observation means.
- A final conclusion written in full sentences.

4. Detailed step-by-step practical procedure

Step	What the student must do	Where / how to do it	What to record and why it matters
1	Create a working folder on the laptop before you begin the practical work.	Open File Explorer. Open Documents. Right-click in an empty space. Choose New > Folder. Type Unit1_Evidence and press Enter.	Record the folder name in your notebook. This matters because every screenshot and note for the lab must be stored in one place so that evidence is not lost.
2	Turn on the personal hotspot on your phone and identify the	If using Android, open Settings > Network & Internet / Connections > Hotspot & tethering / Mobile	Write down the date, your physical location, the phone

	wireless network that you yourself have created for the lab.	Hotspot. If using iPhone, open Settings > Personal Hotspot. Look for the following fields on that screen: network name or device name, Wi-Fi password, band if shown, and security type if shown. Do not use a school network, café network, or another person's hotspot.	model, and the hotspot name. The date and location identify when and where the observation was made. The phone model explains why menu names may differ slightly between devices. The hotspot name is the authorised network for this lab.
3	Open the Wi-Fi list on the laptop and take a screenshot of the wireless environment before connecting.	On Windows, select the Wi-Fi / network icon in the bottom-right corner of the taskbar. A list of visible wireless networks will open. Keep that list on the screen. Press Windows + Shift + S and capture the full list. Save the image as U1_Step3_WiFi_List.png inside Unit1_Evidence.	Record the names of at least three visible networks exactly as shown, without trying to join them. This matters because the screenshot proves that many wireless networks can share the same physical space and may use different security modes.
4	Connect the laptop to your own hotspot only.	In the Wi-Fi list, select your own hotspot name. Choose Connect. Enter the Wi-Fi password shown on the phone. Wait until Windows shows Connected, secured. If Windows asks whether the device should be discoverable on this network, choose No because this is safer for a temporary hotspot environment.	Record whether the connection succeeded on the first attempt. This matters because connection status confirms that the authorised network is working and can now be analysed.
5	Use Command Prompt to retrieve the technical wireless details instead of guessing them from memory.	Press Windows + R. Type cmd and press Enter. In Command Prompt type: netsh wlan show interfaces Press Enter. Look for the fields labelled Name, Description, GUID, Physical address, State, SSID, BSSID, Radio type, Authentication, Cipher, Channel,	Copy the following values into a table in your notebook: SSID, radio type, authentication, cipher, and channel. These values matter because they identify the WLAN standard in use, the protection

		Receive rate, and Transmit rate. Take a screenshot of this result and save it as U1_Step5_Interface_Details.png.	method being used, and the channel on which the network is operating.
6	Interpret the wireless values that you collected so that the lab is analytical and not only descriptive.	Use the interface details screen from Step 5. If the radio type contains values such as 802.11n, 802.11ac, or 802.11ax, classify the connection as a WLAN standard. If the authentication field shows WPA2-Personal, WPA3-Personal, or Open, copy it exactly. If the cipher field shows CCMP or GCMP, record it exactly. If the channel is a low number such as 1 to 13, the connection is likely in the 2.4 GHz range; if it is higher, it may be in the 5 GHz range depending on the device.	Write one short interpretation sentence for each value. Example: 'Authentication is WPA2-Personal, which is stronger than open access because it requires a password.' This matters because the course requires explanation of WLAN standards and security controls, not only screenshots.
7	Observe a personal area network example so that you can distinguish a WLAN from a PAN.	On the phone, open Settings > Bluetooth. On the laptop, open Settings > Bluetooth & devices. Do not pair with another person's device. Simply note that Bluetooth is a short-range personal area network technology used for nearby personal devices. If Bluetooth is off, you may leave it off after checking the menu.	Write two differences between the hotspot WLAN and Bluetooth PAN. For example, note that the hotspot gives Internet sharing over a larger area, while Bluetooth is meant for very short-range personal connections. This matters because Unit 1 includes both WLAN and personal area networks.
8	Harden the hotspot by changing weak or revealing settings on your own device.	Return to the hotspot settings screen on the phone. Check whether the hotspot name reveals your personal name, registration number, or phone number. If it does, rename it to a neutral name such as CYS331_U1_Device. Check the Wi-Fi password. If it is short or easy to guess, change it to a stronger password of at least 12 characters using letters, numbers, and a symbol.	Create a before-and-after table showing the hotspot name, password strength description, and security type. This matters because Unit 1 covers securing WLANs and countermeasures.

		If the phone shows a choice of security types, keep the strongest available option such as WPA2 or WPA3. Take a before screenshot and an after screenshot if the device allows the password field to be hidden safely.	
9	Reconnect and verify that the hardened wireless settings still work correctly.	If the hotspot name or password changed, reconnect the laptop using the new details. Open the Wi-Fi list again, connect to the new hotspot name, and repeat the command netsh wlan show interfaces. Take a final screenshot and save it as U1_Step9_Final_Verification.png.	Record whether the final connection shows Connected, secured and whether the authentication value remained strong. This matters because a security setting is useful only if it is both safer and functional.
10	Prepare the evidence for submission in a logical order.	Open Microsoft Word. Create a report with the headings: Introduction, Device Used, Baseline Observation, Wireless Details, WLAN versus PAN, Security Improvements, Conclusion. Insert the saved screenshots in the order they were taken. Under each screenshot, write a short caption stating what the image shows.	Your report must contain at least four labelled screenshots, one before-and-after configuration table, and a short conclusion. This matters because marks are awarded for both correct practical work and clear evidence.

5. Assessment guide

The following mark distribution applies to this lab. The student earns marks for actual evidence, correct technical interpretation, and professional presentation.

Assessment component	Expected evidence	Marks
Correct setup and baseline evidence	Working folder created, hotspot created, Wi-Fi list screenshot captured, and authorised network clearly identified.	1
Technical observation and interpretation	Correct recording of SSID, radio type, authentication, cipher, channel, plus short explanations of what the values mean.	1

Security hardening and professional report	Before-and-after settings shown, final verification completed, screenshots labelled, and conclusion written clearly.	1

6. Submission checklist

- ☐ A title page with student name, student number, course code, unit title, and date.
- ☐ At least four clearly labelled screenshots saved from the actual practical work.
- ☐ A table containing SSID, radio type, authentication, cipher, and channel.
- ☐ A short comparison paragraph explaining WLAN versus PAN.
- ☐ A before-and-after configuration table showing the security improvement made.
- ☐ A concise conclusion explaining the most important improvement achieved.

7. Common problems and practical fixes

- If the hotspot does not appear in the Wi-Fi list, turn hotspot off and on again, then refresh the laptop Wi-Fi list.
- If netsh wlan show interfaces returns no wireless data, first confirm that the laptop is connected to the hotspot.
- If the screenshot notification disappears, open Snipping Tool from the Start menu and capture the screen again.

8. Final reminder

Do not submit unlabelled screenshots, unexplained values, or copied theory without practical evidence. Every value included in the report must be traceable to a real screen, menu, command output, or diagram produced during your own work.